

NONCLINICAL REPORT

Nonclinical Report No. MOD-4112.1273

Evaluation of in vitro expression of SARS-CoV-2 mRNA and in vivo expression of mRNA-1273

SPONSOR:

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139
Telephone: 617-714-6500

PERFORMING LABORATORY:

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

DATE

29 January 2021

TABLE OF CONTENTS

1. SUMMARY	6
2. DESCRIPTION OF TEST PROCEDURES	8
2.1. Conduct of Study	8
2.1.1. Sponsor	8
2.1.2. Performing Laboratory	8
2.1.3. Study Number	8
2.1.4. Purpose of the Study	8
2.1.5. Institutional Animal Care and Use Committee Approval	8
2.1.6. Study Director	8
2.1.7. Key Personnel	9
2.1.8. Dates of Study Performance	10
2.1.9. Records Maintained	10
2.2. Test Articles	10
2.2.1. mRNA-1273 Test Article	11
2.2.2. Phosphate-Buffered Saline	11
2.3. Test System	12
2.3.1. Evaluation of In Vitro Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen	12
2.3.2. Evaluation of In Vivo Expression of mRNA-1273	12
2.4. Methods	12
2.4.1. Study Design: Evaluation of In Vitro Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen	12
2.4.2. Study Design and Treatment Groups: Evaluation of In Vivo Expression of mRNA-1273	12
3. RESULTS	16
3.1. In Vitro Expression of mRNA encoding the SARS-CoV-2 S-2P Antigen	16
3.1.1. In Vitro Dose Titration of mRNA encoding the SARS-CoV-2 S-2P Antigen	16
3.1.2. Evaluation of In Vitro Cell Surface Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen	17
3.2. In Vivo Expression of mRNA-1273	18
3.2.1. Evaluation of In Vivo Expression of mRNA-1273	18
4. CONCLUSIONS	23
5. REFERENCES	24
6. REPORT APPROVAL	25
7. APPENDICES	26
Appendix 1: SUMMARY OF ANALYSIS: LOT AMPDP-200005 RELEASE	27

LIST OF TABLES

Table 1:	Test Materials and Summary of Dose Formulation Analysis Results	11
Table 2:	Treatment Regimen for Study MOD-4112.1273	13
Table 3:	Complete Fluorescence-Activated Cell Sorting Panel	15

LIST OF FIGURES

Figure 1:	SARS-CoV-2 S-2P Cell Surface Expression MFI*Frequency at 24, 48, and 72 hours With CR3022 Staining.....	16
Figure 2:	In Vitro Cell Surface Expression of the SARS-CoV-2 S-2P Antigen After Transfection of HEK293T Cells With mRNA Encoding SARS-CoV-2 S-2P, Measured by Flow Cytometry at 24 and 48 Hours	17
Figure 3:	In Vivo Expression in Splenic Conventional Dendritic Cells	19
Figure 4:	In Vivo Expression in Splenic Plasmacytoid Dendritic Cells	20
Figure 5:	In Vivo Expression in Lymph Node Conventional Dendritic Cells	21
Figure 6:	In Vivo Expression in Lymph Node Plasmacytoid Dendritic Cells	22

LIST OF ABBREVIATIONS

Abbreviation	Definition
cDC	conventional dendritic cell
CoV	coronavirus
dLN	draining lymph node
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
FACS	fluorescence-activated cell sorting
FBS	fetal bovine serum
GMP	Good Manufacturing Practice
IACUC	Institutional Animal Care and Use Committee
Ig	immunoglobulin
IM	intramuscular(ly)
LNP	lipid nanoparticle
mAb	monoclonal antibody
MFI	mean fluorescent intensity
mRNA	messenger RNA
NTD	N-terminal domain
PBS	phosphate-buffered saline
pDC	plasmacytoid dendritic cell
PEG	polyethylene glycol
PEG2000-DMG	1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000
RBD	receptor binding domain
S	spike
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS-CoV-2	2019 novel coronavirus
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6(undecyloxy)hexyl) amino)octanoate
UTR	untranslated region

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized spike (S) protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic. The safety, immunogenicity, and efficacy of mRNA vaccines against SARS-CoV-2 have been demonstrated in animal challenge studies and in humans in Phase 1, 2, and 3 clinical trials of ModernaTX, Inc.'s mRNA-1273 ([Corbett et al 2020a](#), [Corbett et al 2020b](#), [Jackson et al 2020](#)) and Pfizer Inc.'s BNT162b2 ([Vogel et al 2020](#), [Walsh et al 2020](#)).

The objectives of Study MOD-4112.1273 were to evaluate the in vitro expression of SARS-CoV-2 S-2P-encoding mRNA and the in vivo expression of mRNA-1273.

ModernaTX, Inc. (Sponsor) is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes the SARS-CoV-2 S-2P. The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to the Sponsor's mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (PEG2000-DMG), respectively.

In vitro expression was evaluated in HEK293T cells transiently transfected with a dose titration (0.003125 µg to 0.2 and 0.05 to 0.2 µg) of mRNA encoding SARS-CoV-2 S-2P. After 24, 48, and 72 hours, the cells were collected and assessed for expression of the encoded antigens.

In vivo expression was evaluated in BALB/c mice after they were injected IM (intramuscularly) with 2 or 10 µg of mRNA-1273. Assessments were performed on immune cell populations harvested from spleens and lymph nodes of BALB/c mice 24, 48, and 72 hours after injection. Antibodies specific to the N-terminal domain (NTD, 4A8) and receptor binding domain (RBD, CR3022) of the S protein were evaluated in plasmacytoid dendritic cells (pDCs) and conventional dendritic cells (cDCs) collected from both tissues.

In the evaluation of in vitro expression with a dose titration of mRNA encoding SARS-CoV-2 S-2P, sustained levels of the SARS-CoV-2 S-2P antigens were seen in HEK293T cells over 24, 48, and 72 hours. After transfection with mRNA encoding SARS-CoV-2 S-2P, there was an overall increase in the frequency of cells that express the SARS-CoV-2 S-2P antigen, evident after staining

with mAbs that binds to the RBD (CR3022) and NTD (4A8) epitopes of SARS-CoV-2 S-2P, and the mean fluorescent intensity (MFI) of expression was similar, with a slight dose effect observed.

In the evaluation of in vitro cell surface expression of SARS-CoV-2 S-2P-encoding mRNA, an overall increase in the frequency of cells expressing antigen was observed after staining with CR3022 and 4A8. Evaluation of in vivo expression of mRNA-1273 constructs in the spleen and lymph nodes showed no major differences in RBD or NTD epitope expression, or in antigen-presenting cell expression levels at the similar dose levels. Higher doses drove higher levels of antigen expression in all cell populations measured.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.2. Performing Laboratory

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.3. Study Number

Study number MOD-4112.1273

2.1.4. Purpose of the Study

The objectives of Study MOD-4112.1273 were to evaluate the in vitro expression of SARS-CoV-2 S-2P-encoding mRNA and the in vivo expression of mRNA-1273.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved and conducted in accordance with the regulations of the Sponsor's Institutional Animal Care and Use Committee (IACUC Protocol 18-03-04).

2.1.6. Study Director

(b) (6)

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.7. Key Personnel

Role	Study MOD-4112.1273
Study coordinator	(b) (6) ModernaTX, Inc.
Experimental Design	(b) (6) ModernaTX, Inc.
Operational Activities & Execution	(b) (6) ModernaTX, Inc.
In Vivo Tissue Processing	(b) (6) ModernaTX, Inc.
Flow Analysis	(b) (6) ModernaTX, Inc.
Statistical Analysis	(b) (6) ModernaTX, Inc.
Report compilation:	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Performance

Schedule of Events	Study MOD-4112.1273	
	In Vitro Evaluation	In Vivo Evaluation
Study Initiation	24 Aug 2020 (Study Day 0)	21 Sep 2020 (Study Day 0)
Transfection/injection	24 Aug 2020 (Study Day 0)	21 Sep 2020 (Study Day 0)
Cell/tissue collection	24 hours, 48 hours, 72 hours NA NA	NA 24 hours, 48 hours 24 hours, 48 hours

Abbreviations: NA = not applicable

2.1.9. Records Maintained

Data are archived in the Sponsor Electronic Notebook System at the following location:
Electronic Notebook Project

Data are archived in the Sponsor Electronic Notebook System: Moderna Informatics Digital In Vivo Studies Application

2020_10_27-012- (b) (6)

2.2. Test Articles

The test article used in the in vitro assays was unformulated pre-clinical mRNA encoding the SARS-CoV-2 S-2P antigen. The test article used in the in vivo study was of the mRNA-1273 Drug Product (formulated in LNPs, see below). Phosphate-buffered saline was used as a negative control in the in vivo study.

The preclinical mRNA encoding the SARS-CoV-2 S-2P antigen was manufactured by the Sponsor. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine (Nelson et al 2020). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4), buffer exchanged (b) (4)

sterile filtered, and kept frozen at -20°C until further use. The prepared mRNA was evaluated for expression in HEK293T cells without encapsulation in LNPs.

2.2.1. mRNA-1273 Test Article

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102; cholesterol; DSPC; and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in these studies was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with a same method as the Good Manufacturing Practice mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a novel LNP-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes the full-length SARS-CoV-2 S-2P. mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)). (b) (4)

(b) (4)

(b) (4)

The mixture was neutralized with tris(hydroxymethyl)aminomethane-hydrochloride, pH 7.5; sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273-formulated LNP and stored frozen at –70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin content, and bioburden; the material was deemed acceptable for the in vivo study ([Corbett et al 2020a](#)).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No.	mRNA Lot No.	Size (nm)	PDI	EE%
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; SARS-CoV-2 = 2019 novel coronavirus.

Note: All mRNA test material stocks were formulated in SM-102-containing lipid nanoparticles with a final storage buffer in 10.7 mM sodium acetate; 8.7% sucrose; 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at –70°C until further use.

The EE% is the percentage of mRNA (b) (4)

(b) (4)

(b) (4)

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^a Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4)) was purchased commercially.

2.3. Test System

2.3.1. Evaluation of In Vitro Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen

Cells from the HEK293T (b) (4) cell line, that expresses a mutant version of the SV40 large T antigen.

2.3.2. Evaluation of In Vivo Expression of mRNA-1273

Female BALB/c mice (n = 6 per group), approximately 8-9 weeks old, were obtained from Charles River Laboratories International, Inc. (Kingston, NY). Spleen and lymph node cells were processed for use in the in vivo assays.

2.4. Methods

2.4.1. Study Design: Evaluation of In Vitro Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen

HEK293T cells were transiently transfected with mRNA encoding the SARS-CoV-2 S-2P antigen using a (b) (4) -mRNA transfection kit (b) (4) at the concentrations to be tested in the assays (0.003125 µg to 0.2 and 0.05 to 0.2 µg). After a defined incubation period, the cells were collected and resuspended in fluorescence-activated cell sorting (FACS) buffer (1× phosphate-buffered saline [PBS], 2% fetal bovine serum [FBS]). To detect surface protein expression, the cells were stained with a 1:500 dilution of CR3022 (0.96 mg/ml) (b) (4) in FACS buffer for 1 hour, on ice. Thereafter, cells were washed twice in FACS buffer and incubated with Alexa Fluor 647–goat anti-human immunoglobulin (Ig) G (b) (4) in FACS buffer for 30 minutes on ice. (b) (4) aqua fixable stain (b) (4) were used to assess viability. Data acquisition was performed on a (b) (4) (b) (4)™ instrument (b) (4) and analyzed by FlowJo™ software v.10.

2.4.2. Study Design and Treatment Groups: Evaluation of In Vivo Expression of mRNA-1273

Spleens and lymph nodes were harvested from BALB/c mice and the immune cells from spleens and the draining lymph nodes (dLNs) were isolated for evaluation by flow cytometry. Mice were injected IM with 2 or 10 µg of mRNA-1273 to allow for the evaluation of the in vivo expression of antigen in these immune cells using mAbs specific to the NTD (4A8) or RBD (CR3022). A flow cytometry panel was used to allow for the identification of pDCs and cDCs, and expression levels of the encoded antigens in these cells were measured at 24, 48, and 72 hours. Assays were also performed using immune cells collected from mice immunized with PBS, and the same staining panel was used. The treatment regimen is summarized in [Table 2](#).

Table 2: Treatment Regimen for Study MOD-4112.1273

Group	Number of Animals	Test Article	Total Dose (µg)	Injection	Sample Collection (hours post-Day 0)	Endpoint Monitoring
1	4	PBS	NA	Day 0	24, 48, 72	Cell isolation and flow cytometry
2	4	PBS	NA			
3	6	mRNA-1273	2			
4	6	mRNA-1273	10			

Abbreviations: NA = not applicable; PBS = phosphate-buffered saline.

2.4.2.1. Mouse Spleen Processing Protocol

Splenic cells were transferred into (b) (4) Dissociator C-tubes (b) (4) containing at least 3 mL of media (RPMI 1640 w/ L-Glutamine + 10% FBS + 1% Pen/Strep) and homogenized using protocol m_spleen_01. The contents were transferred to a 15-mL conical tube, centrifuged at 1500 rpm for 5 minutes at 4°C, the supernatant was decanted, and the tube was blotted dry. The cell pellet was resuspended in 3 mL ammonium-chloride-potassium lysing buffer (b) (4) and incubated at room temperature for 5 minutes after which the lysing buffer was quenched with 5 mL of media. The splenocyte resuspension was passed through a 250-µm Pierce cell strainer (b) (4) that was inserted in the tube. The cells were then centrifuged at 1500 rpm for 5 minutes at 4°C, the supernatant decanted, and the cells were resuspended in 2 mL media for use (See [Section 2.4.2.2](#))

2.4.2.2. Lymph Node Processing Protocol

Two lymph nodes were collected from each animal; one lymph node was injected with liberase (b) (4), as described, the other lymph node was not injected.

A 24-well plate was prepared by dispensing 250 µL of RPMI + 2% FBS + 1% Pen/Strep into each well. A fresh solution (within an hour of performing the procedure) of Liberase (250 µg/mL) and DNase (b) (4) (50 µg/mL) was prepared in RPMI + 2% FBS + 1% Pen/Strep. The lymph nodes to be injected were placed in a petri dish and injected with the Liberase/DNase solution using a fine gauge (27 1/2 +) needle, the lymph nodes were swollen. The injected and noninjected lymph nodes were then incubated in the 24-well plate at 37°C for 15 minutes for digestion to occur, followed by the addition of 500 µL RMPI + 10% FBS to quench the digestion. The lymph nodes were then crushed through (b) (4) (70 µm) filters (b) (4) into 15-mL conical tubes and the filter was rinsed with RPMI + 10% FBS. The solution was added to the tubes. The tubes were centrifuged at 1500 rpm for 5 minutes, the supernatant was decanted, and the pellet was resuspended in 200 µL 1× PBS; a small aliquot was taken for counting in a (b) (4) cell counter (b) (4). The suspension was plated in a 96-well round bottom plate and centrifuged at 1500 rpm for 5 minutes.

The splenic suspension ([Section 2.4.2.1](#)) was plated in the same plate and the plating scheme verified. The plates were washed with 1× PBS and centrifuged at 1500 rpm for 5 minutes, followed by staining.

2.4.2.3. In Vivo Expression Staining Protocol

The plates containing the prepared splenic cells ([Section 2.4.2.1](#)) and lymph node cells ([Section 2.4.2.2](#)) were washed once by adding 150 µl of 1× PBS added to each well, centrifuging the plates at 1500 rpm for 5 minutes, and the plate contents discarded (flicked into sink). A 1:1000 aqua viability dye mix (b) (4) was prepared in 1× PBS, 100 µl added to each well and mixed. The plates were then incubated at 4°C for 30 minutes, protected from light, and washed 2 times with 150 µL FACS buffer (1× PBS + 3%FBS). The cells were then resuspended in 50 µL of Fc Block (2.4 G2) diluted at 1:100 in FACS buffer, incubated at 4°C for 5 minutes (protected from light), after which 100 µl of stain mix was added to each well and mixed, and the plates were incubated at 4°C for 30 minutes, protected from light. The plates were washed 2 times with 150 µl 1× FACS buffer, resuspended in 100µL of secondary stain mix (1:500 AL647 anti-human Fc IgG in FACS buffer), and incubated at 4°C for 30 minutes, protected from light. The plates were washed 2 times with 150µL 1× FACS buffer, resuspended in 100 µL (b) (4) for 30 minutes at 4°C, protected from light. Following the incubation with (b) (4), 100 µL 1× PBS was added to each well, and the plates were centrifuged at 1500 rpm for 5 minutes, washed once with 150 µL 1× PBS, then resuspended in 200 µL 1× PBS. The plates were then transferred to an AcroPrep™ 30-40 µm 96-well filter plate and centrifuged at 1500 rpm for 5 minutes. The plates were stored at 4°C, protected from light, and analyzed by flow cytometry as soon as possible.

2.4.2.4 In Vivo Expression Assay Conditions

For the in vivo expression assay, the primary antibodies were CR3022 (RBD-specific) at 1:1600 dilution and 4A8 (NTD-specific) at 1:1600 dilution and the secondary antibody was AL647 anti-human Fc IgG at 1:500 dilution. The cells used in the assays were pDCs and cDCs. The complete FACS panel is shown in [Table 3](#).

Table 3: Complete Fluorescence-Activated Cell Sorting Panel

Target	Fluorophore	Clone
CD169	PerCP/Cy5.5	3D6.112
SIGN-R1	PE	–
IA/IE	AL700	M5/114.152
CD317	BV605	927
Siglec H	BV421	440c
CD3/CD19	FITC	145-2C11/6D5
CD11b	APC/Cy7	M1/70
CD11c	PE/Cy7	HL3
CR3022	AL647	–
4A8	AL647	–

3. RESULTS

3.1. In Vitro Expression of mRNA encoding the SARS-CoV-2 S-2P Antigen

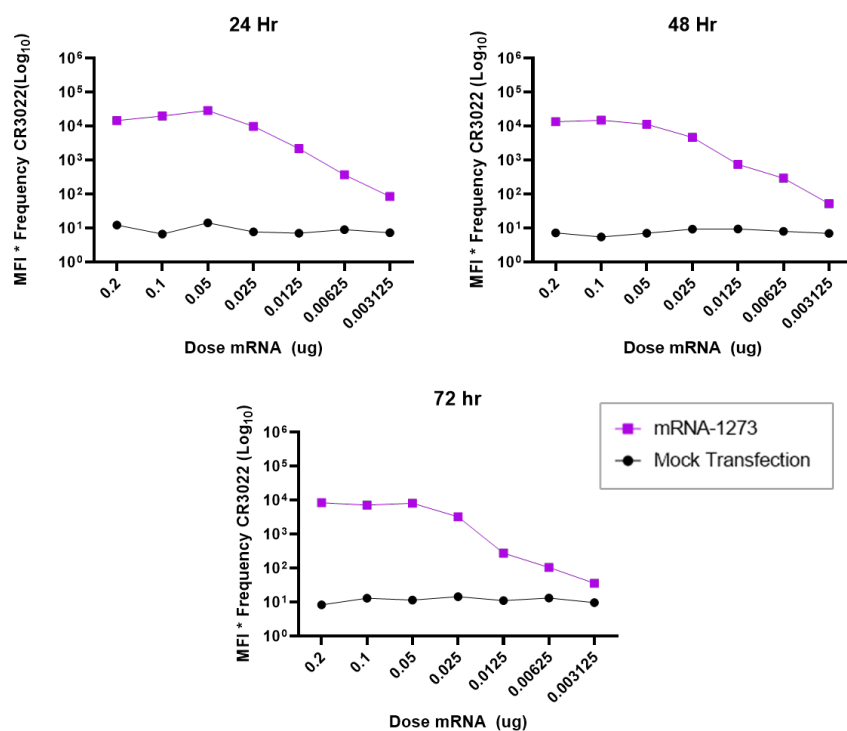
In vitro expression was evaluated in HEK293T cells transiently transfected with mRNA encoding the SARS-CoV-2 S-2P antigen. After 24, 48, and 72 hours, the cells were then collected and assessed for expression of the encoded antigen (as described in [Section 2.4.1](#)).

3.1.1. In Vitro Dose Titration of mRNA encoding the SARS-CoV-2 S-2P Antigen

The cell surface expression was calculated by multiplying the frequency of positive cells by the MFI (MFI $\times$ frequency) and indicates the level of SARS-CoV-2 S-2P antigen expression ([Figure 1](#)).

In vitro expression, with a dose titration of 0.003125 μ g to 0.2 μ g of mRNA, had sustained levels of the SARS-CoV-2 S-2P antigen in HEK293T cells over 24, 48, and 72 hours ([Figure 1](#)).

Figure 1: SARS-CoV-2 S-2P Cell Surface Expression MFI*Frequency at 24, 48, and 72 hours With CR3022 Staining



Abbreviations: MFI = mean fluorescence intensity; MFI*frequency = the frequency of positive cells multiplied by the MFI; mRNA = messenger RNA; SARS-CoV-2 S-2P = 2019 novel coronavirus spike protein with 2 proline substitutions within the heptad repeat 1 domain.

Notes: CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein; in this figure, mRNA-1273 refers to the mRNA construct that encodes the SARS-CoV-2 S-2P.

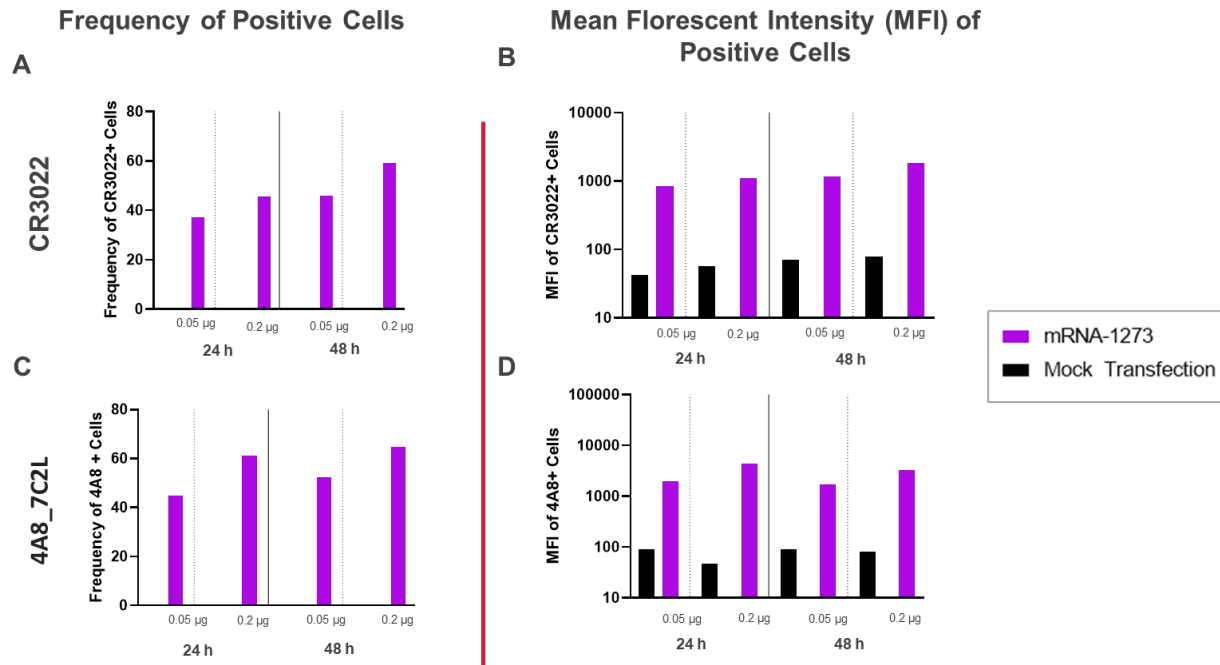
3.1.2. Evaluation of In Vitro Cell Surface Expression of mRNA Encoding the SARS-CoV-2 S-2P Antigen

The frequency and intensity of expression of mRNA encoding SARS-CoV-2 S-2P is shown in Figure 2.

In vitro evaluation of cell surface expression in cells transfected with a dose concentration of 0.05 µg and 0.2 µg mRNA revealed an overall increase in the frequency of cells that express the SARS-CoV-2 S-2P antigen. This was observed after cell staining with mAbs, which bind to the RBD (CR3022) and NTD (4A8) (Figure 2A and Figure 2C). A slightly higher frequency of cells expressing the SARS-CoV-2 S-2P antigen was observed at 48 hours when compared to 24 hours with both CR3022 and 4A8 mAbs (Figure 2A and Figure 2C).

The MFI of expression was similar in cells transfected with SARS-CoV-2 S-2P-encoding mRNA at 24 and 48 hours, evident after staining with CR3022 and 4A8, and a slight dose effect was observed (Figure 2B and Figure 2D). Cells that did not undergo transfection had little to no change over time or with doses.

Figure 2: In Vitro Cell Surface Expression of the SARS-CoV-2 S-2P Antigen After Transfection of HEK293T Cells With mRNA Encoding SARS-CoV-2 S-2P, Measured by Flow Cytometry at 24 and 48 Hours



Abbreviations: MFI = mean fluorescence intensity; mRNA = messenger RNA; SARS-CoV-2 S-2P = 2019 novel coronavirus spike protein with 2 proline substitutions within the heptad repeat 1 domain.

Notes: 4A8 is a monoclonal antibody for the N-terminal domain epitope of the SARS-CoV-2 spike protein; CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein; in this figure, mRNA-1273 refers to the mRNA construct that encodes the SARS-CoV-2 S-2P.

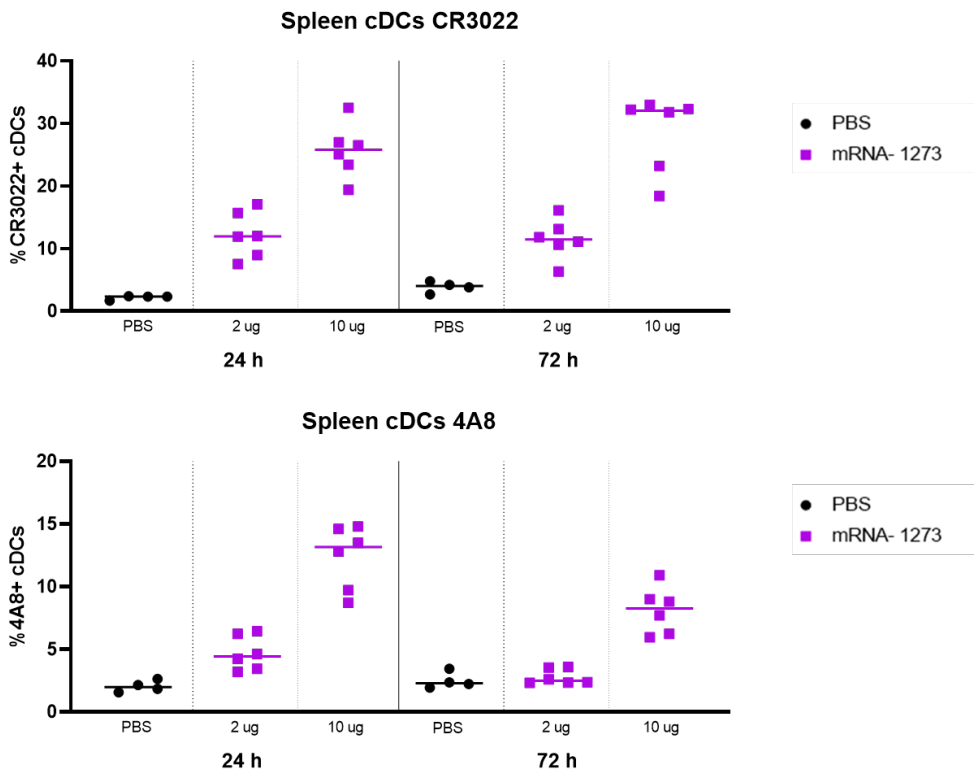
3.2. In Vivo Expression of mRNA-1273

In vivo expression was evaluated in BALB/c mice after they were injected with 2 or 10 µg of mRNA-1273. Assessments were performed on immune cell populations harvested from spleen and lymph nodes from BALB/c mice 24, 48, and 72 hours after injection. Antibodies specific to the NTD of the S protein (4A8) and the RBD of the S protein (CR3022) were evaluated in pDCs and cDCs collected from both tissues. The methods are described in [Section 2.4.2](#).

3.2.1. Evaluation of In Vivo Expression of mRNA-1273

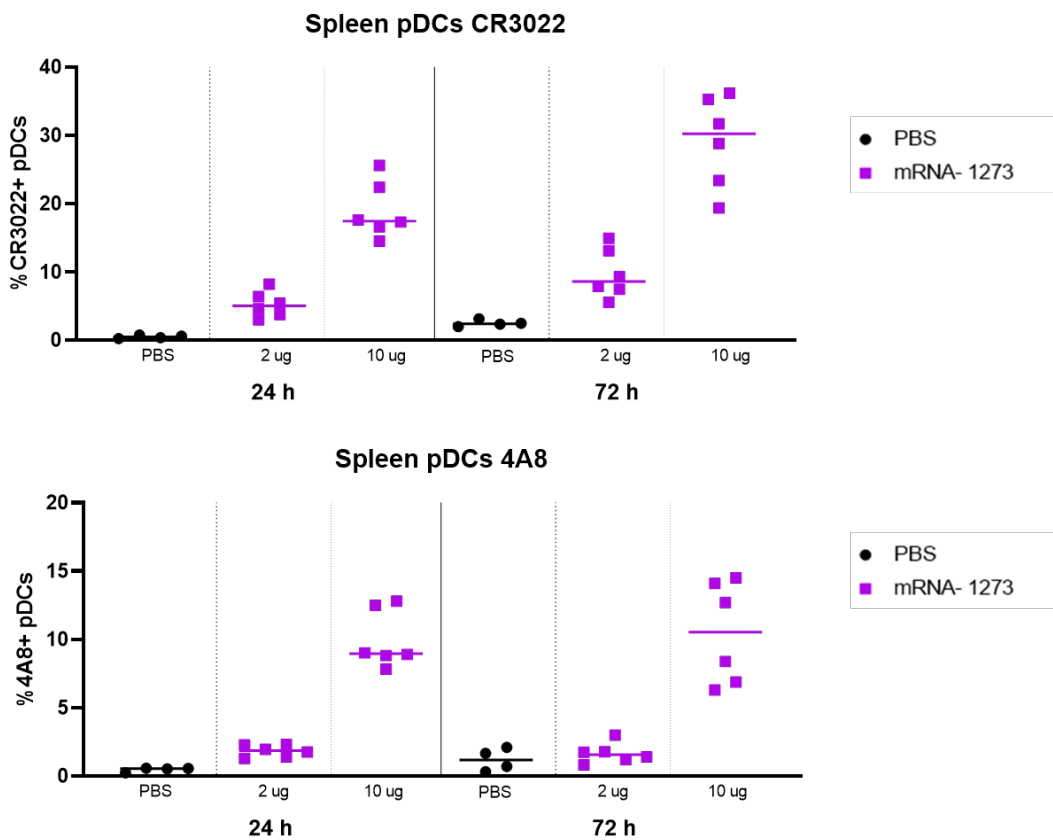
The in vivo expression of mRNA-1273 measured by flow cytometry using 4A8 and CR3022 mAbs in splenic cDCs and pDCs is shown in [Figure 3](#) and [Figure 4](#), respectively, and in dLN cDCs and pDCs in [Figure 5](#) and [Figure 6](#), respectively.

Evaluation of in vivo expression of mRNA-1273 showed very little differences in the level of antigen expression measured in cDCs ([Figure 3](#)) or pDCs ([Figure 4](#)) collected from spleens between 24 hours and 72 hours, regardless of whether the staining was specific to the NTD (4A8 mAb) or RBD (CR3022 mAb) domains. The higher dose drove higher levels of antigen expression in immune cells collected from spleens at each time point.

Figure 3: In Vivo Expression in Splenic Conventional Dendritic Cells

Abbreviations: cDC = conventional dendritic cell; PBS = phosphate-buffered saline.

Notes: 4A8 is a monoclonal antibody for the N-terminal domain epitope of the SARS-CoV-2 spike protein; CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein.

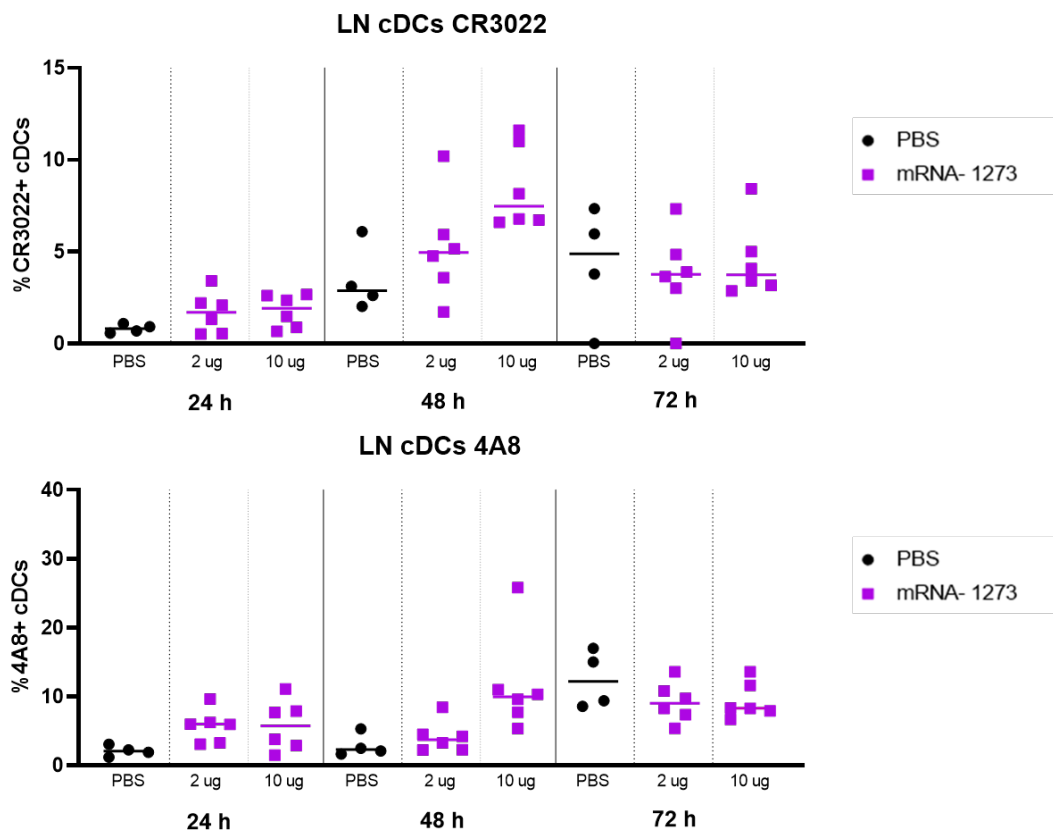
Figure 4: In Vivo Expression in Splenic Plasmacytoid Dendritic Cells

Abbreviations: PBS = phosphate-buffered saline; pDC = plasmacytoid dendritic cell.

Notes: 4A8 is a monoclonal antibody for the N-terminal domain epitope of the SARS-CoV-2 spike protein; CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein.

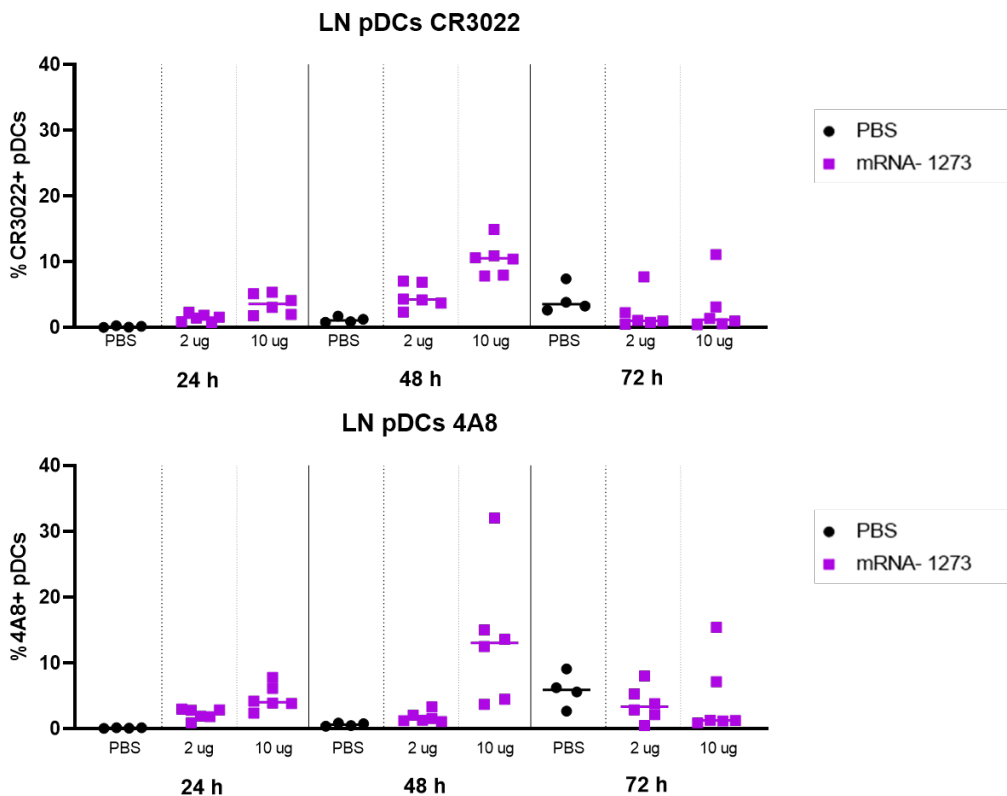
In vivo antigen expression of mRNA-1273 seemed to peak at 48 hours post-immunization in the dLNs both in cDCs (Figure 5) and pDCs (Figure 6) with either the NTD (4A8 mAb) or RBD (CR3022 mAbs).

Figure 5: In Vivo Expression in Lymph Node Conventional Dendritic Cells



Abbreviations: cDC = conventional dendritic cell; LN = lymph node; PBS = phosphate-buffered saline.

Notes: 4A8 is a monoclonal antibody for the N-terminal domain epitope of the SARS-CoV-2 spike protein; CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein.

Figure 6: In Vivo Expression in Lymph Node Plasmacytoid Dendritic Cells

Abbreviations: LN = lymph node; PBS = phosphate-buffered saline; pDC = plasmacytoid dendritic cell.

Notes: 4A8 is a monoclonal antibody for the N-terminal domain epitope of the SARS-CoV-2 spike protein; CR3022 is a monoclonal antibody for the receptor binding domain epitope of the SARS-CoV-2 spike protein.

4. CONCLUSIONS

In the evaluation of in vitro expression with a dose titration (0.003125 µg to 0.2 µg) of mRNA encoding SARS-CoV-2 S-2P, sustained levels of the SARS-CoV-2 S-2P antigens were seen in HEK293T cells over 24, 48, and 72 hours. After transfection with mRNA encoding SARS-CoV-2 S-2P, there was an overall increase in the frequency of cells that express the SARS-CoV-2 S-2P antigen, evident after staining with mAbs that binds to the RBD (CR3022) and NTD (4A8) epitopes of SARS-CoV-2 S-2P, and the MFI of expression was similar, with a slight dose effect observed.

In the evaluation of in vitro cell surface expression of SARS-CoV-2 S-2P-encoding mRNA (0.05 to 0.2 µg), an overall increase in the frequency of cells expressing antigen was observed after staining with CR3022 and 4A8. Evaluation of in vivo expression of mRNA-1273 constructs in the spleen and lymph nodes showed no major differences in RBD or NTD epitope expression, or in antigen-presenting cell expression levels at the similar dose levels. Higher doses drove higher levels of antigen expression in all cell populations measured.

5. REFERENCES

Corbett KS, Edwards D, Leist SR, Abiona OM, Boyoglu-Barnum S, Gillespie RA et al. SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness. *Nature*. 2020a Aug 5;10.1038/s41586-020-2622-0. Epub 2020 Aug 5.

Corbett KS, Flynn B, Foulds KE, Francica JR, Seyhan B-B, Werner AP, et al. Evaluation of the mRNA-1273 vaccine against SARS-CoV-2 in nonhuman primates. *N Engl J Med* 2020b;383:1544-55.

Hassett, KJ, Benenato KE, Jacquinet E, Lee A, Woods A, Yuzhkov O, et al. Optimization of Lipid Nanoparticles for Intramuscular Administration of mRNA Vaccines. *Mol Ther Nucleic Acids*. 2019 Apr 15;15:1-11.

Jackson LA, Anderson EJ, Rouphael NG, Roberts PC, Makhene M, Coler RN, et al. An mRNA vaccine against SARS-CoV-2 – preliminary report. *N Engl J Med*. 2020 Nov 12;383(20):1920-31.

Nelson J, Sorensen EW, Mintri S, Rabideau AE, Zheng W, Besin G, et al. Impact of mRNA chemistry and manufacturing process on innate immune activation. *Sci Adv*. 2020 Jun 24;6(26):eaaz6893.

Vogel AB, Kanevsky I, Che Y, Swanson KA, Muik A, Vormehr M, et al. A prefusion SARS-CoV-2 spike RNA vaccine is highly immunogenic and prevents lung infection in non-human primates. *bioRxiv*. 2020:2020.09.08.280818. Epub 2020 Sep 8.

Walsh EE, Frenck RW Jr, Falsey AR, Kitchin N, Absalon J, Gurtman A, et al. Safety and immunogenicity of two RNA-based Covid-19 vaccine candidates. *N Engl J Med*. 2020:NEJMoa2027906. Epub ahead of print 2020 Oct 14.

6. REPORT APPROVAL

Study Director:

See esignature and date signed on last page of document

(b) (6)

ModernaTX, Inc.

Date

7. APPENDICES

Appendix 1 Summary of Analysis: Lot AMPDP-200005 Release

APPENDIX 1: SUMMARY OF ANALYSIS: LOT AMPDP-200005 RELEASE



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

Test	Method	Target Attributes		Results
Lipid Identification:	DPTM-0019	(b) (4)		
SM-102				
Cholesterol				
DSPC				
PEG2000-DMG				
Lipid Content (mg/mL)				
SM-102	DPTM-0019	(b) (4)		(b) (4)
Cholesterol				
DSPC				
PEG2000-DMG				
Lipid Impurities (% Area)				
Individual Impurities	DPTM-0019	(b) (4)		
Total Impurities				
pH	USP <791> DPTM-0028			
Osmolality	USP <785> DPTM-0029			
Bacterial Endotoxin	USP <85>			
Bioburden	USP<61>			
Particulate Matter	USP <788> Method 2			



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval

Signature Page for VV-NC-000876 v2.0

Approval	(b) (6) 29-Jan-2021 15:24:33 GMT+0000
----------	--

Signature Page for VV-NC-000876 v2.0